Journal of Energy Storage 72 (2023) 108727

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Research papers

Experimental assessment and comparison of single-phase versus two-phase liquid cooling battery thermal management systems

Nan Wu [a,*], Yisheng Chen [b], Boshen Lin [b], Junjie Li [b], Xuelong Zhou [b,*]

[a] School of Mechatronic Engineering and Automation, Foshan University, Foshan 528200, China
[b] Shenzhen Key Laboratory of New Lithium-ion Batteries and Mesoporous Materials, College of Chemistry and Environmental Engineering, Shenzhen University, Shenzhen 518060, China

ARTICLE INFO

Keywords:
Battery thermal management
Liquid cooling
Two-phase
Experimental
Large-format cell

ABSTRACT

Liquid-based cooling systems are becoming the dominant approach for thermal management of lithium-ion batteries due to the favorable specific heat capacity and heat transfer coefficient. In this study, single-phase and two-phase liquid cooling (SPLC and TPLC) systems are experimentally evaluated and compared in two indirect-contact modes for a large-format lithium-ion battery. Two commonly used liquid coolants, e. g. Novec 7000 and deionized water, are utilized as heat transfer fluids. The experimental results reveal that both SPLC and TPLC systems with coolant flow exhibit excellent ability to control temperature spikes, and the SPLC system using deionized water outperforms the TPLC system using Novec 7000 in terms of cooling capability. Additionally, the cooling effect of copper foam configuration surpasses that of straight fin configuration. The SPLC system equipped with copper foam holds the best cooling performance, with a maximum surface temperature of 32.1 °C and a largest temperature difference of 1.8 °C observed over the 3C discharge process at a flow rate of 10 mL/min. These findings are valuable for designers seeking to develop efficient and practical thermal management systems for large-format lithium-ion batteries.

1. Introduction

With the increasingly prominent energy crisis and environmental problems, countries worldwide are actively formulating energy development strategies and exploring renewable energy sources. Lithium-ion battery has become the prevailing choice for powering electric vehicles and hybrid electric vehicles due to its high specific energy density, long cycle life and no memory effect. Without doubt that the battery determines the power performance of the whole vehicle, however its performance degradation and safety issues remain significant are still major obstacles to the widespread application of electric vehicles [1–4].

As an electrochemical power source, lithium-ion battery will generate considerable heat during the charging and discharging processes due to joule heating and entropy change [4–6]. Moreover, the temperature sensitivity of lithium-ion batteries is pronounced, with recommended operating temperatures typically ranging between 25 °C and 40 °C [7]. Differences in reaction current density, electrical resistance and the external dissipative environment lead to the increased inconsistencies in heat and power distribution within a single battery unit or among battery packs. Rapid degradation of the battery cell comes with inconsistent temperatures. The synergistic effect of large temperature peaks and inconsistent temperatures will greatly accelerate the premature aging of lithium-ion batteries [8–10]. Therefore, it's imperative to implement an efficient battery thermal management system [11–19] to avoid excessive temperature rise and temperature imbalance, and ultimately extend the service life while improving safety.

Liquid-based battery thermal management (BTM) systems have become the dominant strategy due to their relatively higher specific heat capacity and superior heat transfer coefficient. These systems can be further subcategorized into two modes: indirect-contact and direct-contact. In the indirect-contact mode, the heat transfer fluid circulates within cooling channels to dissipate heat indirectly from the battery surface. The cooling channels can take the form of metal plates with built-in channels or discrete tubes. Liquid water is the preferred heat transfer fluid in indirect liquid cooling systems owing to its affordability and ease of control [20–23]. On the other hand, direct-contact liquid-based BTM systems, also known as immersion cooling systems, usually rely on dielectric fluids. Candidate fluids for such systems include mineral oils [24,25], silicone oils [26,27], refrigerants [28–30], hydrofluoroethers [31–33], as well as fuels such as propane and ammonia [34–36].

* Corresponding authors.
E-mail addresses: wunan@fosu.edu.cn (N. Wu), xzhoual@szu.edu.cn (X. Zhou).

https://doi.org/10.1016/j.est.2023.108727
Received 20 February 2023; Received in revised form 8 August 2023; Accepted 12 August 2023
Available online 24 August 2023

Nomenclature

H_{lv}	latent heat of evaporation, J/kg
m	mass, kg
Q	heat flow, J
T	temperature, K
ΔT	temperature difference, K
α	effective evaporation ratio of coolant

Subscripts

c	coolant
eff	effective
i	temperature measurement point
in	inlet
l	liquid phase
max	maximum
min	minimum
out	outlet

Abbreviations

BTM	battery thermal management
CF	copper foam
DI	deionized
NAC	natural air cooling
SF	straight fin
SPLC	single-phase liquid cooling
TPLC	two-phase liquid cooling

In recent years, there has been a growing interest in employing direct liquid cooling systems for BTM utilizing hydrofluoroethers like Novec engineered fluids. These hydrofluoroether products manufactured by 3M possess favorable boiling points and latent heat of evaporation. Researchers have extensively studied and developed immersion cooling systems using Novec for cylindrical batteries and large-format pouch cells. For example, Hirano et al. [37] designed a two-phase immersion cooling system using Novec 7000 and Novec 649 (with boiling points of 34 °C and 49 °C respectively) to cool a battery module consisting of ten series-connected cylindrical cells. They conducted thorough performance evaluations of the module under aggressive conditions at 10C and 20C. In our previous works [38,39], we introduced a phase-change immersion cooling system using Novec 7000 for a 20 Ah pouch cell and executed experimental tests under various operating conditions. The results indicated that the temperature spikes and gradients of the battery could be successfully controlled within the desired range by adjusting the coolant quantity and flow modes. In addition, An et al. [40] examined a mini-channel liquid cooling system using Novec 7000 instead of immersing the battery module in coolant, and the battery temperature could be finely controlled around 40 °C.

The above literature review suggests the potential of Novec-based liquid cooling systems in regulating battery temperatures, offering promising solutions for efficient thermal management in various battery configurations. However, the practical implementation of such systems encounters several challenges including increased complexity, cost and weight. In contrast, indirect-contact cooling systems that utilize water as the heat transfer fluid offer easier implementation with lower weight. Additionally, water exhibits higher specific heat capacity and thermal conductivity compared to Novec 7000. Liquid-based BTM systems, in general, prove effective in managing battery temperatures, although the choice of system and fluid depends on specific application requirements. Nonetheless, the lack of sufficient experimental data poses a challenge in establishing a consensus on the preferred type of fluid or contact mode for deployment.

This study aims to evaluate the effectiveness of indirect-contact, single-phase and two-phase liquid cooling systems (SPLC and TPLC) for large-format lithium-ion batteries through a comparative experimental approach. Two commonly used liquid coolants, e. g. Novec 7000 and deionized (DI) water, are employed as the heat transfer fluids. Moreover, two cooling configurations involving straight fins and copper foam are designed for the SPLC and TPLC systems. The impact of critical parameters on the thermal behaviors of the battery is analyzed and disclosed. The discussions and findings from this research will provide valuable reference for designers seeking efficient and practical thermal management systems for large-format lithium-ion batteries.

2. Experimental

2.1. System description

The indirect-contact liquid cooling BTM system with straight fin configuration is designed, and its schematic diagram is exhibited in Fig. 1(a) along with the exploded views. The pouch battery used in this work is the same as the one tested in our previous research on direct immersion cooling, with technique specifications summarized in Table 1. As shown in Fig. 1(a), the battery is sandwiched between two 6061 aluminum alloy cold plates via a 0.3 mm thick thermal interface material (TIM), i.e. the silica gel pad. The dimensions of the base plate are 146 mm × 220 mm × 2 mm, while the cold plate features a fin pitch of 12.5 mm, fin thickness of 1.2 mm and fin height of 4 mm. To enclose the coolant chamber, a polycarbonate plastic cover is joined to the base with several fastening bolts. To prevent coolant leakage, a custom-made O-sealing ring is placed inside the chamber, and the external joint is sealed using 704 room temperature vulcanized silicone rubber from Liyang kangda new material Co., Ltd., China. One picture of the specimen with straight fins is presented in Fig. 1(c).

Meanwhile, another indirect-contact liquid cooling mode incorporating copper foam is also considered to enhance convective and heat transfer, which has the same material and base plate dimensions as the straight fin cold plate above. The copper foam with dimensions of 204 mm × 128 mm × 4 mm is inserted into the empty coolant chamber. The copper foam blocks have a pore density of 20 PPI and a high open porosity larger than 98 %.

Based on the above literature review, deionized (DI) water and 3M's dielectric engineered fluid Novec 7000 are selected as the representatives of single-phase and two-phase liquid coolants for a comparative analysis with different BTM structure designs. Compared to the engineered fluid Novec 7000, liquid water has three times the specific heat capacity and an order of magnitude greater thermal conductivity, allowing for efficient heat absorption and heat dissipation. Despite that, the dielectric engineered fluid wins in other ways due to the key features such as excellent thermal insulation, environmentally friendly, good material compatibility and non-flammable, which has been widely used in a variety of heat transfer applications in the semiconductor, electronics and chemical manufacturing industries. In addition, the boiling point is suitable for the ideal operating temperature range of lithium-ion batteries to effectively utilize the latent heat. Therefore, this investigation aims to assess and compare the cooling performance of the two liquid coolants with different BTM designs. The key properties of the two coolants are tabulated in Table 2.

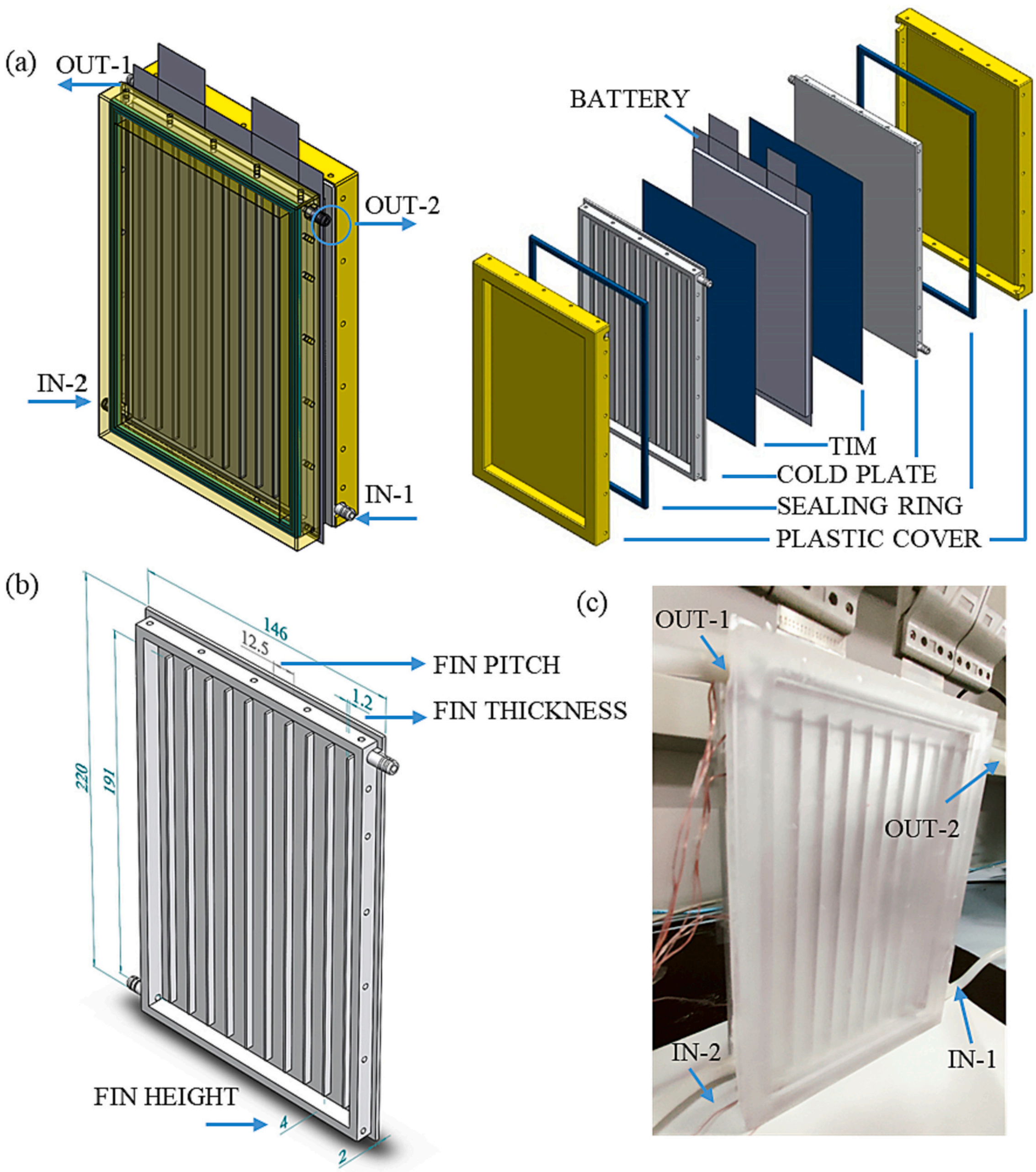


Fig. 1. (a)Structure and exploded views of the liquid cold plate with straight fins; (b) dimensions of the single cold plate; (c) photo of the specimen with straight fins.

Table 1
Main technique specifications of the pouch cell.

Items	Specifications
Cathode material	$LiFePO_4$
Anode material	Graphite
Standard capacity	20 Ah
Dimensions	142 mm*240 mm*8.3 mm
Rated voltage	3.2 V
Maximum charge voltage	3.65 V
Cut-off voltage	2 V
Specific energy	130 Wh/kg

Table 2
Physical properties of the two coolant types.

Coolant	DI water	Novec 7000
Boiling point, @1 atm (°C)	100	34
Liquid density, kg/m^3	997 (25 °C)	1400
Liquid thermal conductivity, W/(m K)	0.609	0.075
Liquid specific heat, kJ/(kg K)	4.179	1.300
Latent heat of vaporization, kJ/kg	2260	142.0
Surface tension, mN/m	72.8	12.4

2.2. Experimental setup

As depicted in Fig. 2. the experimental setup in this work is mainly composed of the battery testing system, the cooling system and the data acquisition system. The multi-channel battery testing station (BTS-20V100A, Shenzhen Neware Electronic Co. Ltd., China) with an accuracy of ±0.1 % FS is used to control the charging and discharging process of the tested battery. The operational parameters, such as current-voltage curves, are collected and stored in real time by a monitoring computer. The high-precision peristaltic pump (Masterflex L/S®, Easy-Load II) is used to supply coolant to the BTM system, and the volume flow rate of liquid coolant is adjustable with an accuracy of ±1 %. T-type thermocouples (TT-T-30-SLE, OMEGA) with an accuracy of ±0.4 % are employed to measure the point temperatures of the battery surface.

As shown in Fig. 3, six temperature measurement points are arranged according to the general distribution characteristics of the battery surface temperature, which have been tested with the IR images in the battery in our previous studies. It is widely recognized that the temperature is higher near the tabs than the bottom of the battery core, especially at high discharge rates. Hence, the six representative measuring points, i.e. near the positive tab, near the negative tab, at the center and at the bottom, are selected to monitor the temperature spike and temperature gradient. The temperature data are recorded by a

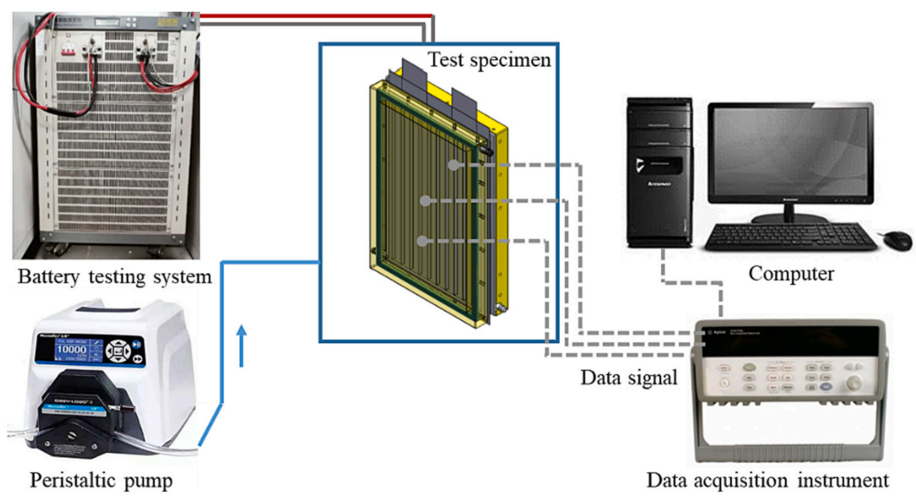


Fig. 2. Schematic diagram of experimental system.

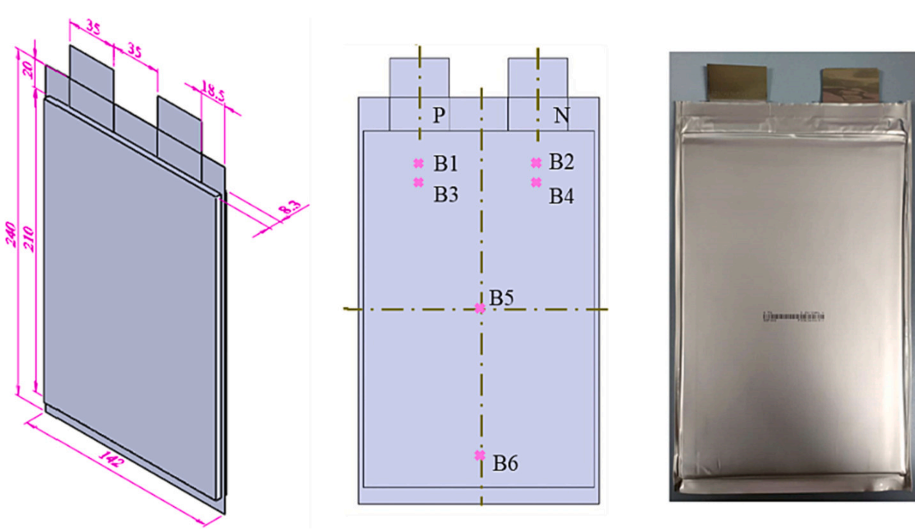


Fig. 3. Dimensions of the battery and arrangement of temperature measurement points.

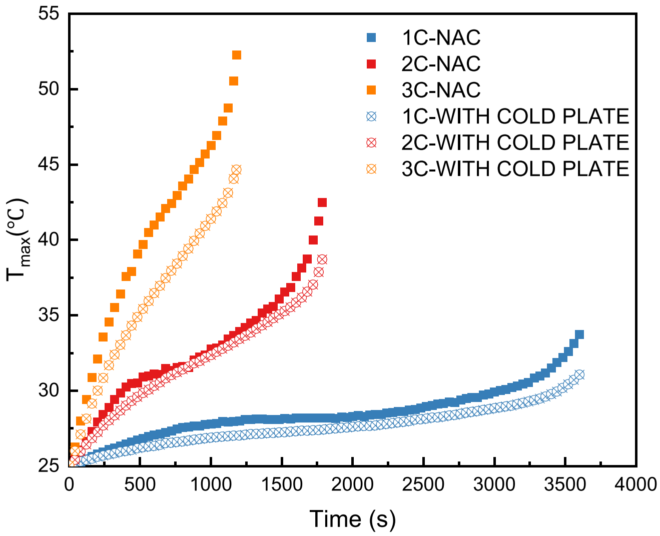


Fig. 4. The evolution of T_{max} with time for the two passive cooling cases at different discharge rates.

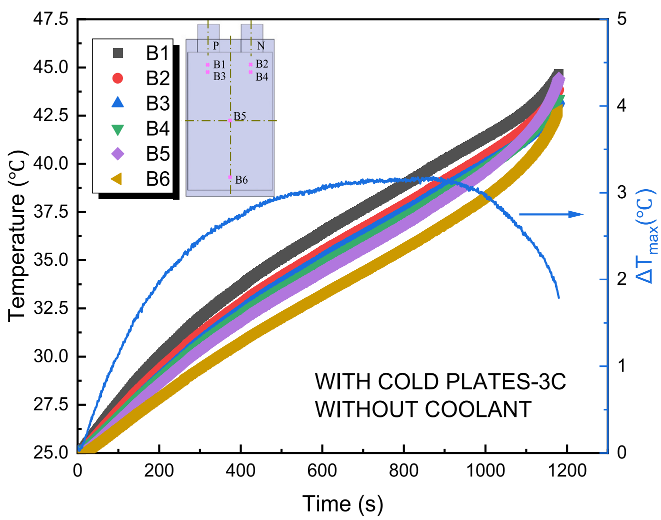
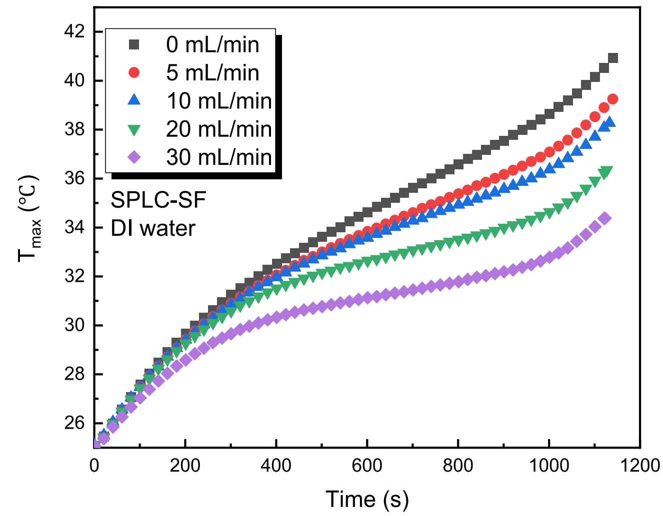
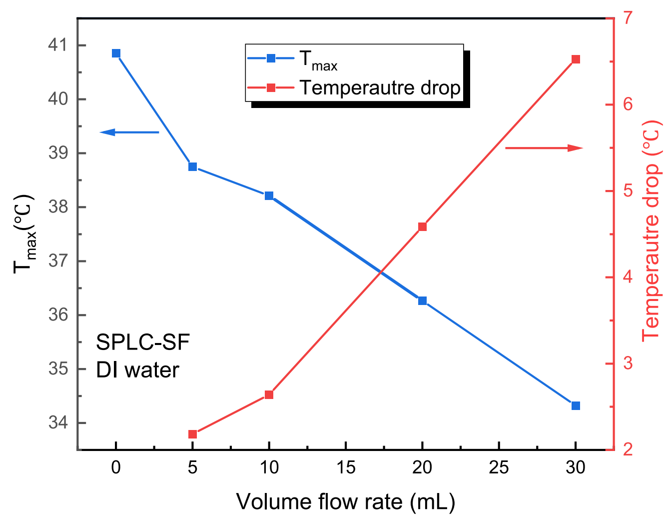


Fig. 5. Variations of battery surface temperature and temperature difference for the passive cooling with cold plate at 3C discharge rate.



(a)



(b)

Fig. 6. SPLC-SF system with coolant flow at 3C discharge rate: (a) T_{max} with volume flow rate; (b) T_{max} and temperature drop under various flow rates.

multi-channel data logger (Agilent, 34970A) with an acquisition card (Agilent, 34901A) and then transmitted to the monitor computer. All the experiments are conducted in lab under an ambient temperature of 25 °C. The inlet coolant temperature and initial battery temperature remain unchanged and are consistent with the ambient temperature. By analyzing the elevated battery temperature with and without coolant flow, the cooling performance brought by the coolant can be indirectly judged to a certain extent.

The battery is mainly tested under the discharge process, as studies have found that the generated heat in the discharge process is more intense than the charge process at the same current flow. The battery is discharged under the constant current mode at different discharge rates of 1C, 2C and 3C with a cut-off voltage of 2 V. C-rate is defined as the current through the battery divided by the theoretical current under which the battery would deliver its nominal rated capacity in one hour, e.g. 2C indicates the current that can fully charge or discharge a battery in 0.5 h. According to the specifications provided by the battery manufacturer (TeamGiant New Power Co., Ltd., China), the continuous discharge current of the tested pouch cell should not exceed 3C (60A), thus the maximum discharge rate is set to 3C in this work. The thermal performance of the two proposed indirect-contact cooling configurations are evaluated and compared by adjusting the volume flow rates of the liquid coolants, i.e. DI water and Novec 7000. For convenience, the liquid cooling using deionized water is called SPLC (i.e. single-phase liquid cooling), and that using Novec 7000 is TPLC (i.e. two-phase liquid cooling). For the cold plate with straight fins or copper foam, the suffixes SF and CF are added, e.g. SPLC-CF refers to the single-phase liquid cooling system with copper foam.

2.3. Experimental uncertainty

The maximum surface temperature and temperature difference of the battery are usually used as important evaluation parameters to judge the thermal performance. Two critical parameters, i.e. the maximum surface temperature (T_{max}) and temperature difference of the battery (ΔT_{max}), are derived from the point temperatures measured by thermocouples at specific locations on the battery surface as shown in Fig. 3. The two parameters can be obtained by the following formulas:

$$T_{max} = max\{T_i\}, i \text{ varies from 1 to 6} \tag{1}$$

$$\Delta T_{max} = T_{max} - T_{min} \tag{2}$$

where T_{max} and T_{min} represent the maximum and minimum temperatures among the measured points on the battery surface respectively, T_i denotes the point temperature measured by the thermocouples at position B_i.

According to the widely accepted error propagation formula, the calculated maximum relative errors of two indirect parameters were below 2 % and 5 % respectively under the experimental condition in this work.

3. Discussions of experimental results

3.1. Passive cooling without coolant injection

Prior to exploring the cooling characteristics of the liquid-based BTM designs, the temperature rise process of the pouch cell is tentatively tested under the condition of natural air cooling (NAC) and with the adoption of straight fin cold plates (without coolant injection).

Fig. 4 presents the evolution of T_{max} over time for the two passive cooling methods at different discharge rates. It's clear that the battery temperature increases gradually with increasing depth of discharge due to the heat accumulation effect. The temperature rise is greater at higher discharge rates due to additional heat generated, which can be estimated using the available empirical formula, i.e. the Bernardi equation.

At 1C discharge, the battery temperature increase is not significant, and the thermally conductive cold plate without coolant supplementation reduces the T_{max} by 2.6 °C. However, at 2C and 3C discharge rates, the battery temperature rises drastically and more markedly towards the end of the discharge. Here, the cold plate further reduces the T_{max} of the battery by 3.8 °C and 7.6 °C, respectively, as it timely transfers the generated heat to the attached metal plate. Such heat dissipation effect is more obvious at higher discharge rates with larger temperature differences between the battery and surroundings. T_{max} drops to 44.7 °C at 3C for the passive cooling with cold plate. Analysis of temperature variations at six measured points as depicted in Fig. 5 shows a gradually decreasing trend in spatial temperature distribution from top to bottom of the battery, attributing to the dual effects of uneven heat generation rate in different regions and heat transfer form tabs. The battery temperature is generally within the safe operation range under 1C and 2C discharge conditions with the help of the metal base plates. Therefore, testing is done at 3C discharge rates to evaluate the proposed BTM configurations in following experiments.

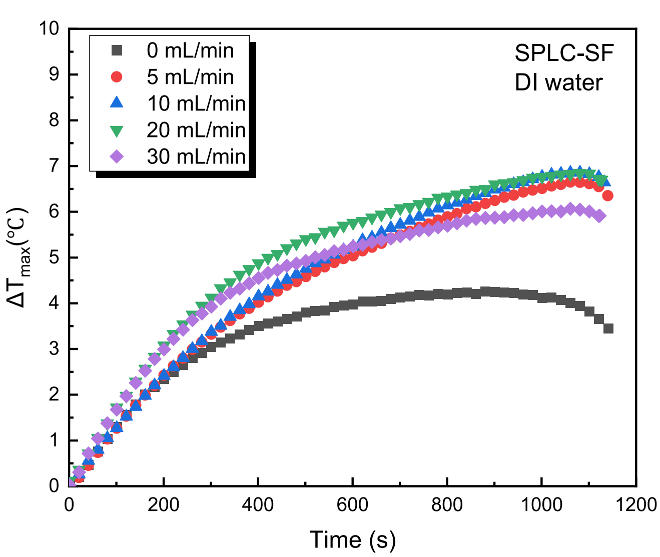


Fig. 7. Maximum temperature difference of the SPLC-SF system at 3C discharge.

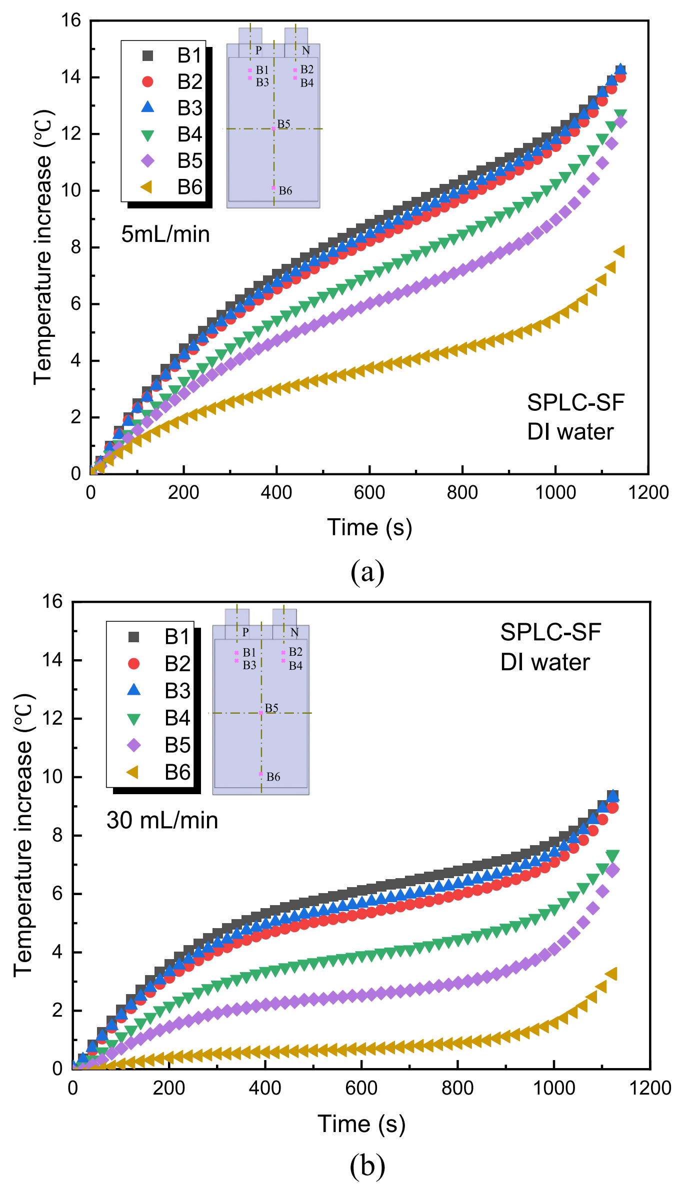


Fig. 8. Temperature rise of the measured points at different flow rates for SPLC-SF system with coolant flow at 3C discharge rate: (a) 5 mL/min; (b) 30 mL/min.

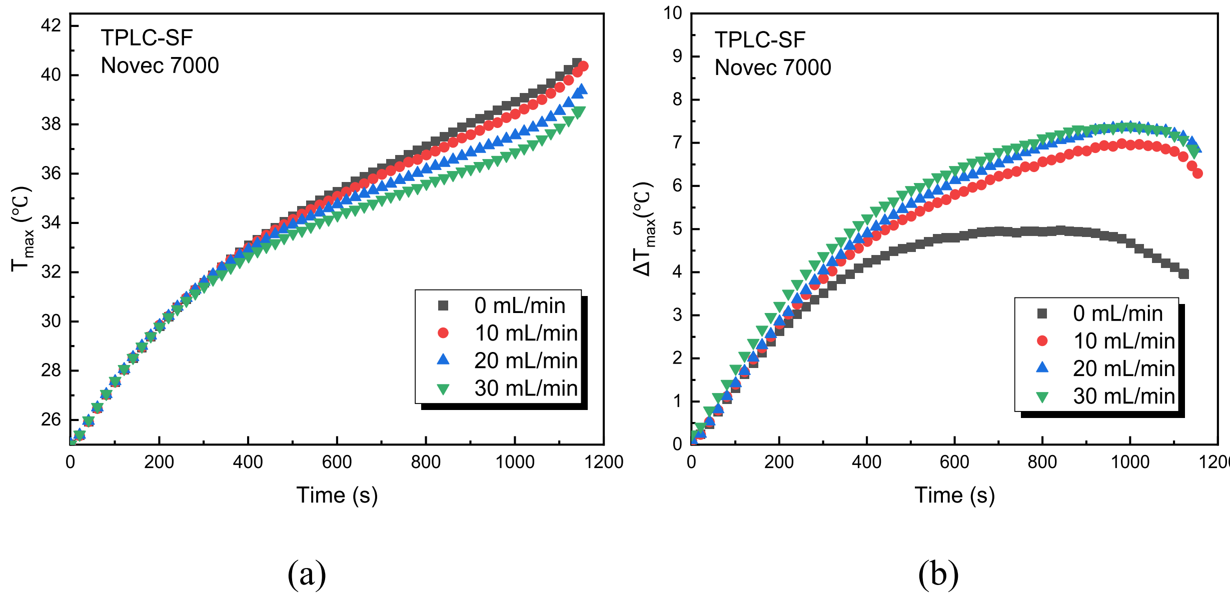


Fig. 9. TPLC-SF system with coolant flow at 3C discharge rate: (a) maximum battery temperature; (b) temperature difference.

3.2. Cooling performance of the cold plate with straight fins

This section examines the performance of the straight fin cold plate using two different coolant media, i.e. deionized (DI) water and Novec 7000, and the effects of coolant flow rates are also disclosed.

3.2.1. Single-phase liquid cooling with straight fin (SPLC-SF)

Fig. 6(a) displays the changing trend of T_{max} with time at different coolant flow rates for the SPLC-SF system. T_{max} further decreases to 40.9 °C when the chamber is filled with liquid coolant compared to the passive cooling only with metal cold plates (without liquid coolant). The coolant flow contributes to a lower T_{max}. As the flow rate increases from 5 mL/min to 30 mL/min, T_{max} is reduced from 38.7 °C to 34.3 °C, but the improvement effect contributed by the coolant per unit mass flow is constantly diminishing. Compared to the passive cooling without coolant injection, the cooling contribution by the coolant flow can be judged by the temperature drop , i.e. the temperature difference of the battery with and without liquid cooling. Fig. 1(b) summaries the total temperature drop brought by the liquid coolant with different volume flow rats. The temperature drop caused by the unit flow rate is about 0.44 °C at a volume flow rate of 10 mL/min, decreasing to 0.23 °C at 30 mL/min. This suggests there may also be a cooling capacity limit for SPLC-SF system. Once a certain volumetric flow rate is reached, increasing flow rates will only increase the pump power without enhancing the cooling effect.

Fig. 7 shows that at the initial moment, the maximum temperature difference (ΔT_{max}) under different flow rates is not significant. While at the end of discharge, the influence of flow rate is more obvious, i.e. the maximum temperature difference between measurement points is the smallest in the static mode (i.e. 0 mL/min), and it first increases and then decreases with the increasing coolant flow rate. The larger temperature gradient in the flow mode is mainly caused by the dual effect of coolant temperature change in the flow direction and heat generation difference in the battery. In flow mode, the bottom of the battery is generally colder when coolant is injected from below, which can be confirmed by the variations of temperature rise at different flow rates in Fig. 8. At a flow rate of 5 mL/min, the temperature rise is 7.9 °C at the bottom and 14.2 °C at the top, decreasing to 3.3 °C and 9.4 °C respectively at 30 mL/min.

3.2.2. Two-phase liquid cooling with straight fin (TPLC-SF)

Fig. 9 depicts the changes in surface temperature and temperature difference at different flow rates for the TPLC system using Novec 7000 fluid. The experimental results indicate that the battery temperature is insensitive to the increase in flow rate. Specifically, T_{max} is about 40.5 °C in static mode and the value remains high at 38.6 °C even when the flow rate is raised to 30 mL/min. The additional temperature drop caused by the coolant flow from 10 mL/min to 30 mL/min is only 1.8 °C. The limited cooling effect of TPLC-SF system at high flow rates may be caused by the lower specific heat capacity and thermal conductivity of the Novec 7000 fluid. In addition, the flow of Novec 7000 fluid also affects the temperature uniformity of the battery. The temperature uniformity is best at rest mode with a maximum temperature difference of 5.0 °C. The temperature difference gradually increases with the increase in flow rate under the test conditions. The ΔT_{max} is 7.4 °C at a flow rate of 30 mL/min.

Actually, the effective heat flow brought away by the coolant can be broadly estimated according to the energy balance equation of the coolant.

$$Q_{eff} = m_c c_{p,l}\left(T_{c,out} - T_{c,in}\right) + \alpha m_c H_{lv} \tag{1}$$

where Q_{eff} is the effective heat flow applied to the coolant media, m_c is the mass amount of coolant, $T_{c,out}$ and $T_{c,in}$ are the outlet and inlet temperatures of the coolant, respectively, $c_{p,l}$ is the specific heat of the coolant in liquid state, α is the effective evaporation ratio of coolant, H_{lv} is the latent heat of evaporation. Eq. (1) also explains why the conclusion that the two-phase cooling media is more effective than single-phase cooling media is conditional.

As listed in Table 2, the specific heat capacity of Novec 7000 fluid is only one-third of that of deionized water, which means that the effective sensible heat removed by liquid water will be three times that of Novec 7000 fluid at the same temperature rise according to Eq. (1). The cooling performance of SPLC system is superior to that of the TPLC system in the indirect contact mode with straight fins, as illustrated in Figs. 6–10. Fig. 10(b) presents the cooling power of the two types cooling systems, which is determined by dividing the temperature drop of the battery by the mass flow rate of the coolant. At the same volume flow rate of 10 mL/min, the temperature drop per unit mass flow for the SPLC system is

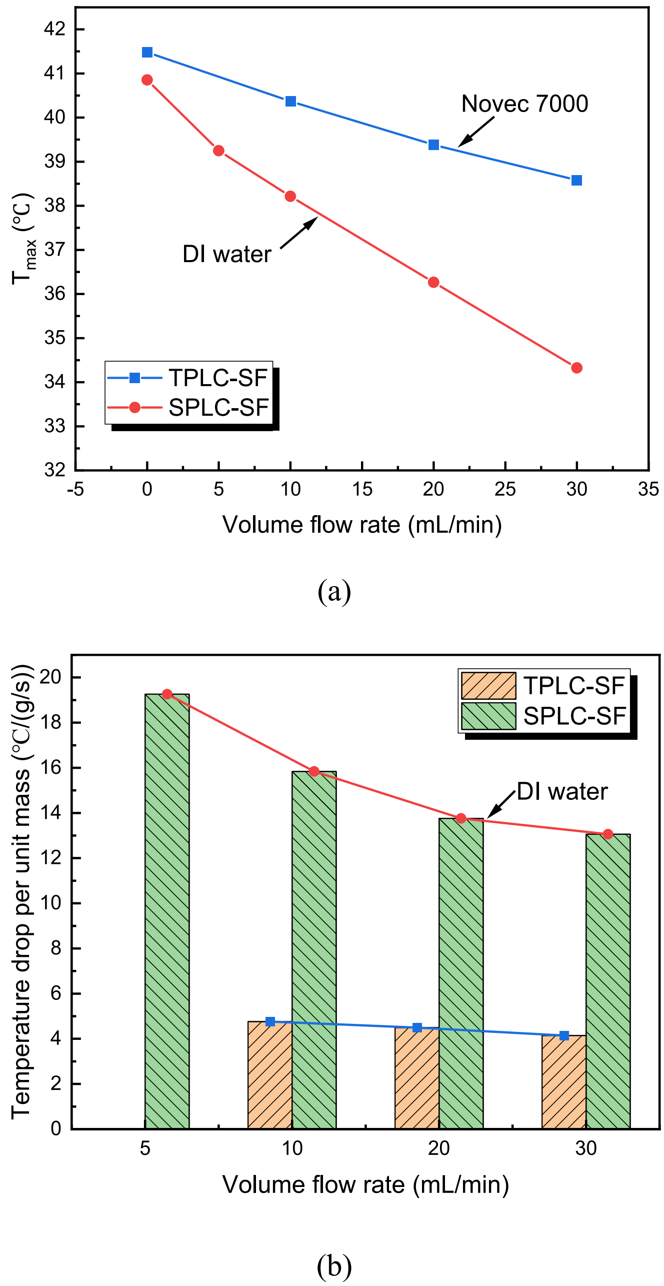


Fig. 10. Comparison of SPLC-SF and TPLC-SF: (a) maximum battery temperature; (b) temperature drop contributed by unit mass flow rate of coolant.

15.8 °C/(g/s), while it is only 4.8 °C/(g/s) for the TPLC system. In general, the cooling capacity of the SPLC system using DI water is outstanding in the indirect contact mode with straight fin structure.

3.3. *Cooling performance of copper foam filled cold plate*

Another series of experiments are carried out to evaluate the cooling performance of indirect-contact liquid cooling using copper foam configuration. Figs. 11 and 12 compare the transient changes in battery surface temperature and temperature difference of the SPLC-CF and TPLC-CF systems during the 3C discharge process. When the flow rate increases from 0 mL/min to 30 mL/min, the maximum battery temperature of the SPLC-CF system decreases from 36.8 °C to 28.3 °C, while the maximum battery temperature of the TPLC-CF system changes only 1.7 °C and remains as high as 35.3 °C. The results indicate that the battery surface temperature of the SPLC-CF system is more sensitive to the change of volume flow rate than the TPLC-CF system due to the higher specific heat capacity of water coolant, which is similar to the straight fin structures as shown in Fig. 10.

Moreover, whether single-phase or two-phase cooling media, the cooling effect and temperature uniformity of the copper foam structure is much better compared to the straight fin configuration. In other words, due to the massive open pore structures and high thermal conductivity of the copper foam, the cooling performance of SPLC-CF and TPLC-CF is superior to the SPLC-SF and TPLC-SF, respectively. As depicted in Fig. 13, the temperature difference is well controlled below 2.5 °C for the SPLC-CF and TPLC-CF systems, which is far below the 5 °C criteria. For example, at a volume flow rate of 10 mL/min, the SPLC-CF system provides a maximum surface temperature and a largest temperature difference of 32.1 °C and 1.8 °C, respectively. In summary, the above assessment and comparison suggest that, in the indirect contact mode, the single-phase liquid cooling system with copper foam (SFLC-CF) has the best cooling performance in simultaneously regulating the temperature spikes and temperature gradient of the large-format lithium-ion battery.

These findings can aid in optimizing system configurations and coolant selections to enhance the overall thermal performance of such batteries. However, in realistic applications, batteries need to be connected in the form of packs or modules and the cooling requirements are more intense. Cooling plate systems expanded to the battery pack need to be designed according to the size and shape of the battery pack. Multiple cooling plates can be connected to fit the overall size of the battery pack and increase its cooling surface area. Besides, the above findings indicate that whether single phase or two-phase cooling media, only in specific heat transfer structures and application scenarios can they be maximally effective, achieving a balance between thermal management effectiveness and cost. In our future work, we intend to conduct in-depth comprehensive researches to figure out this problem with more professional experimental set-up and the intervention of numerical simulations.

4. Conclusions

In this work, the indirect-contact single-phase and two-phase liquid cooling systems (SPLC and TPLC) are explored for a large-format lithium-ion battery through a comparative experimental study. Two typical heat transfer fluids, e. g. Novec 7000 and deionized water, are employed for the SPLC and TPLC systems, respectively. In addition, two types of cooling configurations with straight fins and copper foam are also conceived. Through analysis and discussion, the flowing conclusions can be drawn:

i. The sole natural air cooling system can't meet the cooling requirements at high discharge rate of 3C, and the maximum battery temperature can be lowered by 7.6 °C by using passive cooling with cold plate.
ii. In the indirect contact mode, the cooling effect of the SPLC system using deionized water is superior to that of the TPLC system using Novec 7000. At a volume flow rate of 10 mL/min, the cooling power of SPLC system is approximately three times that of the TPLC system.
iii. Compared with the straight fin structure, the embedded copper foam cooling structure exhibits better cooling performance in regulating temperature spikes and temperature gradients simultaneously.

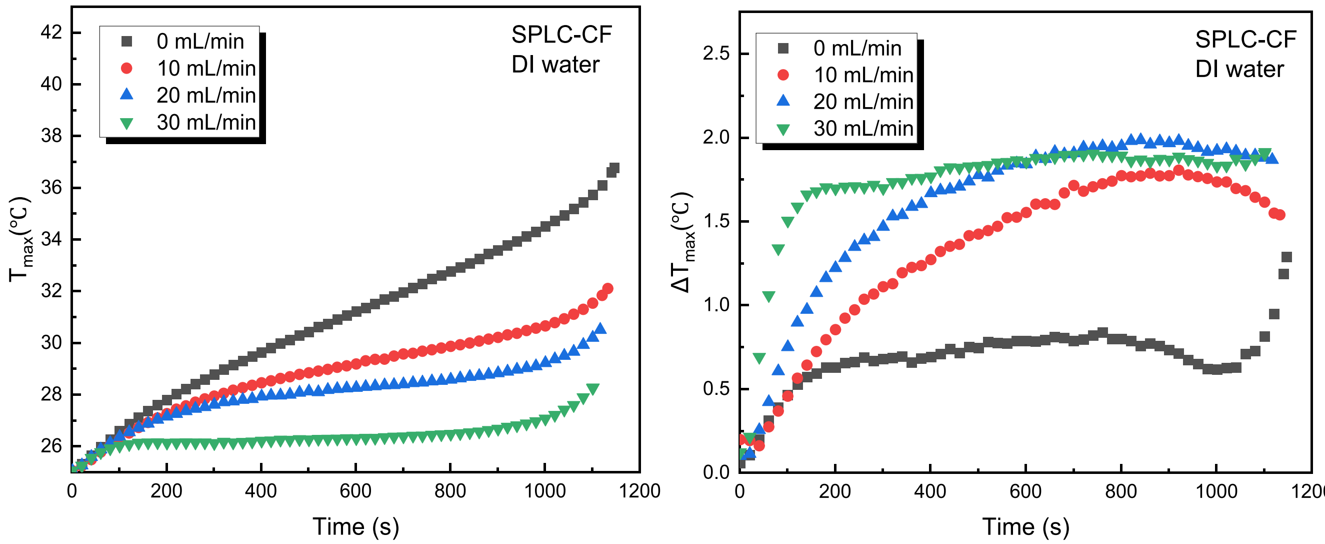


Fig. 11. SPLC-CF system: transient variations of maximum battery temperature and temperature difference at different volume flow rates under 3C discharge.

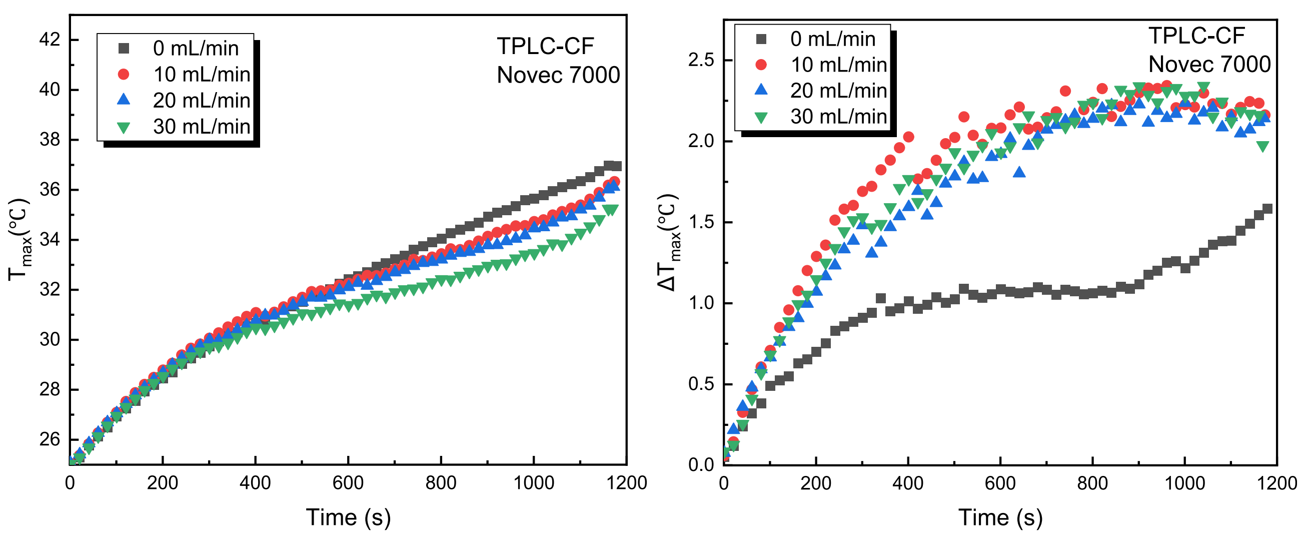


Fig. 12. TPLC-CF system: transient variations of maximum battery temperature and temperature difference at different volume flow rates under 3C discharge.

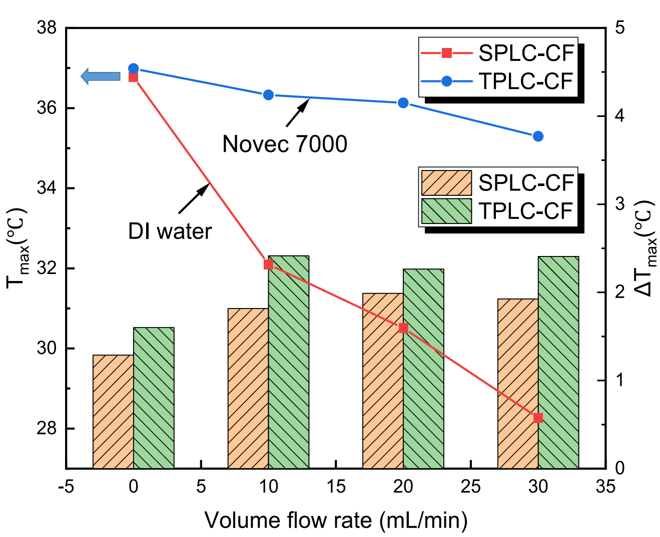


Fig. 13. Comparison of SPLC-CF and TPLC-CF: maximum battery temperature and temperature difference.

iv. The SPLC system with copper foam demonstrates the most effective cooling capability, successfully controlling the maximum surface temperature and the largest temperature difference at 32.1 °C and 1.8 °C over 3C discharge process with a volume flow of 10 mL/min.

CRediT authorship contribution statement

Nan Wu: Funding acquisition, Conceptualization, Methodology, Data curation, Writing – original draft, Writing – review & editing. **Yisheng Chen:** Writing – review & editing. **Boshen Lin:** Writing – review & editing. **Junjie Li:** Writing – review & editing. **Xuelong Zhou:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This work was supported by National Natural Science Foundation of China (No. 52006149), Shenzhen Science and Technology Program (No. JCYJ20190808123011253) and Guangdong Basic and Applied Basic Research Foundation (2022A1515140070).

References

[1] D. Ren, X. Feng, L. Lu, X. He, M. Ouyang, Overcharge behaviors and failure mechanism of lithium-ion batteries under different test conditions, Appl. Energy 250 (2019) 323–332.
[2] S. Panchal, I. Dincer, M. Agelin-Chaab, M. Fowler, R. Fraser, Uneven temperature and voltage distributions due to rapid discharge rates and different boundary conditions for series-connected $LiFePO_4$ batteries, Int. Commun. Heat Mass Transf. 81 (2017) 210–217.
[3] W. Song, M. Chen, F. Bai, S. Lin, Y. Chen, Z. Feng, Non-uniform effect on the thermal/aging performance of Lithium-ion pouch battery, Appl. Therm. Eng. 128 (2018) 1165–1174.
[4] E. Schuster, C. Ziebert, A. Melcher, M. Rohde, H.J. Seifert, Thermal behavior and electrochemical heat generation in a commercial 40 Ah lithium ion pouch cell, J. Power Sources 286 (2015) 580–589.
[5] K. Chen, G. Unsworth, X. Li, Measurements of heat generation in prismatic Li-ion batteries, J. Power Sources 261 (2014) 28–37.
[6] J. Zhang, J. Huang, Z. Li, B. Wu, Z. Nie, Y. Sun, F. An, N. Wu, Comparison and validation of methods for estimating heat generation rate of large-format lithium-ion batteries, J. Therm. Anal. Calorim. 117 (1) (2014) 447–461.
[7] A.A. Pesaran, Battery thermal models for hybrid vehicle simulations, J. Power Sources 110 (2002) 377–382.
[8] T. Grandjean, A. Barai, E. Hosseinzadeh, Y. Guo, A. McGordon, J. Marco, Large format lithium ion pouch cell full thermal characterisation for improved electric vehicle thermal management, J. Power Sources 359 (2017) 215–225.
[9] T. Waldmann, G. Bisle, B.-I. Hogg, S. Stumpp, M.A. Danzer, M. Kasper, P. Axmann, M. Wohlfahrt-Mehrens, Influence of cell design on temperatures and temperature gradients in Lithium-ion cells: an in operando study, J. Electrochem. Soc. 162 (6) (2015) A921–A927.
[10] M. Fleckenstein, O. Bohlen, M.A. Roscher, B. Bäker, Current density and state of charge inhomogeneities in Li-ion battery cells with LiFePO4 as cathode material due to temperature gradients, J. Power Sources 196 (10) (2011) 4769–4778.
[11] G. Zhao, X. Wang, M. Negnevitsky, H. Zhang, A review of air-cooling battery thermal management systems for electric and hybrid electric vehicles, J. Power Sources 501 (2021), 230001.
[12] Y. Yang, L. Chen, L. Yang, X. Du, Numerical study of combined air and phase change cooling for lithium-ion battery during dynamic cycles, Int. J. Therm. Sci. 165 (2021), 106968.
[13] Y. Lv, G. Liu, G. Zhang, X. Yang, A novel thermal management structure using serpentine phase change material coupled with forced air convection for cylindrical battery modules, J. Power Sources 468 (2020), 228398.
[14] H. Chen, T. Zhang, Q. Gao, Z. Han, Y. Xu, K. Yang, X. Xu, X. Liu, Advance and prospect of power battery thermal management based on phase change and boiling heat transfer, J. Energy Storage 53 (2022), 105254.
[15] Z. Ling, S. Li, C. Cai, S. Lin, X. Fang, Z. Zhang, Battery thermal management based on multiscale encapsulated inorganic phase change material of high stability, Appl. Therm. Eng. 193 (2021), 117002.
[16] R. Jilte, A. Afzal, S. Panchal, A novel battery thermal management system using nano-enhanced phase change materials, Energy 219 (2021), 119564.
[17] H.M. Jaffal, N.S. Mahmoud, A.A. Imran, A. Hasan, Performance enhancement of a novel serpentine channel cooled plate used for cooling of Li-ion battery module, Int. J. Therm. Sci. 184 (2023), 107955.
[18] X. Wang, T. Ma, Y. Zhu, H. Chen, J. Zeng, Experimental investigation on startup and thermal performance of a high temperature special-shaped heat pipe coupling the flat plate heat pipe and cylindrical heat pipes, Exp. Thermal Fluid Sci. 77 (2016) 1–9.
[19] W. Wu, S. Xie, W. Zhang, R. Ma, J. Yang, Z. Rao, Role of natural convection and battery arrangement for phase change material based battery thermal management unit, J. Energy Storage 52 (2022).
[20] M.S. Patil, J.-H. Seo, S. Panchal, S.-W. Jee, M.-Y. Lee, Investigation on thermal performance of water-cooled Li-ion pouch cell and pack at high discharge rate with U-turn type microchannel cold plate, Int. J. Heat Mass Tran. 155 (2020), 119728.
[21] Z. Chen, S. Yang, M. Pan, J. Xu, Experimental investigation on thermal management of lithium-ion battery with roll bond liquid cooling plate, Appl. Therm. Eng. 206 (2022), 118106.
[22] M. Akbarzadeh, T. Kalogiannis, J. Jaguemont, L. Jin, H. Behi, D. Karimi, H. Beheshti, J. Van Mierlo, M. Berecibar, A comparative study between air cooling and liquid cooling thermal management systems for a high-energy lithium-ion battery module, Appl. Therm. Eng. 198 (2021), 117503.
[23] Q.L. Yue, C.X. He, H.R. Jiang, M.C. Wu, T.S. Zhao, A hybrid battery thermal management system for electric vehicles under dynamic working conditions, Int. J. Heat Mass Tran. 164 (2021), 120528.
[24] J. Liu, Y. Fan, Q. Xie, Feasibility study of a novel oil-immersed battery cooling system: experiments and theoretical analysis, Appl. Therm. Eng. 208 (2022), 118251.
[25] J. Liu, Y. Fan, J. Wang, C. Tao, M. Chen, A model-scale experimental and theoretical study on a mineral oil-immersed battery cooling system, Renew. Energy 201 (2022) 712–723.
[26] T. Amalesh, N. Lakshmi Narasimhan, Cooling of a lithium ion battery using phase change material with air/dielectric fluid media: a numerical study, Proc. Inst. Mech. Eng., Part A: J. Power Energy 234 (5) (2019) 722–738.
[27] T. Amalesh, N. Lakshmi Narasimhan, Liquid cooling vs hybrid cooling for fast charging lithium-ion batteries: a comparative numerical study, Appl. Therm. Eng. 208 (2022), 118226.
[28] X. Wang, H. Zhang, Z. Yi, X. Huang, H. Guo, S. Zhu, Thermal characteristics of refrigerant flow boiling in two mini-channel heat sinks of different aspect ratios for battery thermal management, Appl. Therm. Eng. 217 (2022), 119173.
[29] M. Shen, Q. Gao, Structure design and effect analysis on refrigerant cooling enhancement of battery thermal management system for electric vehicles, J. Energy Storage 32 (2020), 101940.
[30] Y. Fang, W. Yang, D. Xu, L. Hu, L. Su, Y. Huang, Experimental investigation on flow boiling characteristics of R1233zd(E) in a parallel mini-channel heat sink for the application in battery thermal management, Int. J. Heat Mass Tran. 178 (2021), 121591.
[31] X. Tan, P. Lyu, Y. Fan, J. Rao, K. Ouyang, Numerical investigation of the direct liquid cooling of a fast-charging lithium-ion battery pack in hydrofluoroether, Appl. Therm. Eng. 196 (2021), 117279.
[32] Y.F. Wang, J.T. Wu, Thermal performance predictions for an HFE-7000 direct flow boiling cooled battery thermal management system for electric vehicles, Energy Convers. Manag. 207 (2020), 112569.
[33] R.W. van Gils, D. Danilov, P.H.L. Notten, M.F.M. Speetjens, H. Nijmeijer, Battery thermal management by boiling heat-transfer, Energy Convers. Manag. 79 (2014) 9–17.
[34] M. Al-Zareer, I. Dincer, M.A. Rosen, Development and analysis of a new tube based cylindrical battery cooling system with liquid to vapor phase change, Int. J. Refrig. 108 (2019) 163–173.
[35] M. Al-Zareer, I. Dincer, M.A. Rosen, A novel approach for performance improvement of liquid to vapor based battery cooling systems, Energy Convers. Manag. 187 (2019) 191–204.
[36] M. Al-Zareer, I. Dincer, M.A. Rosen, Development and evaluation of a new ammonia boiling based battery thermal management system, Electrochim. Acta 280 (2018) 340–352.
[37] T.T.H. Hirano, T. Hasegawa, T. Sekiguchi, M. Uchino, Boiling liquid battery cooling for electric vehicle, in: 2014 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC Asia-Pacific), 2014.
[38] N. Wu, Y. Chen, B. Lin, J. Li, X. Zhou, Thermal performance evaluation of boiling cooling system for the high-rate large-format lithium-ion battery under coolant starvations, J. Energy Storage 55 (2022), 105616.
[39] N. Wu, X. Ye, J. Yao, X. Zhang, X. Zhou, B. Yu, Efficient thermal management of the large-format pouch lithium-ion cell via the boiling-cooling system operated with intermittent flow, Int. J. Heat Mass Tran. 170 (2021), 121018.
[40] Z. An, L. Jia, X. Li, Y. Ding, Experimental investigation on lithium-ion battery thermal management based on flow boiling in mini-channel, Appl. Therm. Eng. 117 (2017) 534–543.